Problem Set 4. Causality and Studies
Quantitative Methods for Humanities

1. Efficacy of Small Class Size in Early Education

The STAR (Student–Teacher Achievement Ratio) Project was a four-year longitudinal study investigating the effects of class size in early grades on educational performance and personal development. This study, conducted from 1985 to 1989, involved 11,601 students who were randomly assigned to one of three groups: small classes, regular-sized classes, or regular-sized classes with a teacher's aid. The experiment cost approximately $12 million, and although the program ended in 1989 when the first kindergarten cohort completed third grade, data collection continued through high school.

The original data file, `STAR.csv`, contains the following variables (see Table 1). For this required problem set, use the summary tables below. Students who want extra practice may optionally reproduce the tables from the original data in R, but no code is required for the answers.

Table 1: Description of Variables in the STAR Study

Variable	**Description**
`race`	Student's race (1 = white, 2 = black, 3 = Asian, 4 = Hispanic, 5 = Native American, 6 = others)
`classtype`	Type of kindergarten class (1 = small, 2 = regular, 3 = regular with aid)
`g4math`	Total scaled score for the math portion of the fourth-grade standardized test
`g4reading`	Total scaled score for the reading portion of the fourth-grade standardized test
`yearssmall`	Number of years spent in small classes
`hsgrad`	High-school graduation status (1 = graduated, 0 = did not graduate)

Table 2: Fourth-grade outcomes by kindergarten class type

	Reading score			Math score		
Class type	**n**	**Mean**	**SD**	**n**	**Mean**	**SD**
Small	722	723.52	51.37	722	709.32	43.70
Regular	833	719.73	53.11	833	709.40	41.03
Regular with aid	789	720.81	52.47	789	707.48	44.85

Table 3: Selected test-score quantiles by kindergarten class type

	Reading score		Math score	
Class type	**33rd pct.**	**66th pct.**	**33rd pct.**	**66th pct.**
Small	705	741	694	726
Regular	705	740	696	724
Regular with aid	705	738	696	725

Table 4: Number of years spent in small classes

Kindergarten class type	**0 years**	**1 year**	**2 years**	**3 years**	**4 years**
Small	0	576	272	195	857
Regular	1961	95	58	80	0
Regular with aid	1996	97	60	78	0

Table 5: Outcomes by years spent in small classes

Years	**Read mean**	**Read med.**	**Math mean**	**Math med.**	**Grad. rate**
0	719.91	722	707.97	710	0.829
1	723.04	724	708.06	709	0.791
2	717.87	720	711.14	714	0.813
3	718.62	721	708.90	709	0.832
4	724.91	726	710.10	711	0.878

Table 6: Race gaps in selected STAR outcomes

Class type	**Race**	**n scores**	**Reading mean**	**Math mean**	**n grad.**	**Grad. rate**
Small	Black	113	697.91	697.61	235	0.745
Small	White	605	728.10	711.35	664	0.867
Regular	Black	124	689.35	698.53	288	0.740
Regular	White	705	724.95	711.29	790	0.857

The dataset contains missing values, which occur due to students leaving STAR schools before third grade or joining later. The tasks below analyze the effects of small class sizes.

1. **Treatment and Outcomes:** What is the treatment in the STAR experiment? Which outcomes in Table 1 are short-term outcomes, and which one is a longer-term outcome?

2. **Comparison of Test Scores:** Using Table 2, compare fourth-grade reading and math test scores between students assigned to small classes and those in regular-sized classes. Interpret the results. To assess the size of the effect, compare the mean differences to the standard deviations.

3. **Quantile Treatment Effects:** Using Table 3, compare high scores (66th percentile) and low scores (33rd percentile) for small classes and regular classes. Does this add much beyond the mean comparison?

4. **Effect of Years in Small Classes:** Some students participated in small classes for all four years, while others only experienced one year.

   (a) Using Table 4, describe how years spent in small classes are distributed across kindergarten class types.

   (b) Using Table 5, investigate whether more years in small classes are associated with higher test scores and graduation rates.

5. **Reducing Racial Achievement Gaps:** Using Table 6, compare white and black students within regular-sized classes and within small classes. Did small classes clearly reduce achievement gaps? Provide a substantive interpretation.

6. **Long-Term Effects on Graduation:** Assess whether the STAR program influenced high-school graduation rates.

   (a) Compare graduation rates by years spent in small classes using Table 5.

   (b) Analyze whether the STAR program reduced racial disparities in graduation rates using Table 6.

   (c) What caution should we use when interpreting the relationship between years spent in small classes and later outcomes?

7. **Optional R Extension:** If you want extra coding practice, reproduce one of the tables above from `STAR.csv`. This is optional enrichment and is not required for the problem set.

2. The Anchoring Effect

The **anchoring effect** is a cognitive bias in which individuals' decisions are affected by an initial piece of information, called the *anchor*. This phenomenon was first formally identified and studied by Nobel laureate Daniel Kahneman and his coauthor Amos Tversky in their 1974 paper on judgment under uncertainty.[1]

Originally, Kahneman and Tversky asked participants to estimate the percentage of African countries in the United Nations, after presenting them with a rigged "wheel of fortune" that landed on 10 or 65. In a more recent replication effort (ManyLabs 1 study), participants were asked about the height of Mount Everest. Some were randomly given a "high" anchor, others a "low" anchor, and each participant then provided a numerical estimate of Everest's height. The replication data is contained in the `anchoring.RData` dataset, whose relevant variables are described in Table 7.

[1] Tversky, Amos, and Daniel Kahneman. 1974. Judgment under uncertainty: heuristics and biases. Science 185 (4157): 1124–1131.

Table 7: Key Variables in the `anchoring` Dataset

Variable	Description
`anchor`	The randomly assigned anchor (`low` or `high`)
`everest_feet`	Judged height of Mount Everest in feet
`sex`	Participant's reported sex (`m` or `f`)
`age`	Participant's age in years
`citizenship`	country code of citizenship
`lab_or_online`	Indicator of whether the study was done online or in-lab

Table 8: Randomization balance in the anchoring study

Sample	Anchor	n	Mean age	Female share
In-lab	High	1050	20.33	0.730
In-lab	Low	1147	20.24	0.710
Online	High	1195	29.97	0.610
Online	Low	1240	30.23	0.635

Table 9: Everest-height estimates by anchor

Sample	Anchor	n	Mean estimate	SD
All	High	2245	34607.88	9325.77
All	Low	2387	11510.55	10225.87
In-lab	High	1050	35350.39	9514.13
In-lab	Low	1147	9166.77	8850.19
Online	High	1195	33955.47	9111.22
Online	Low	1240	13678.55	10914.96

Table 10: Anchoring effects in selected countries

Country	n high	n low	High mean	Low mean	Difference
Brazil	35	46	33750.7	9659.0	24091.7
Canada	63	83	34922.4	6302.5	28619.9
India	32	36	28906.8	9947.8	18959.1
Italy	49	58	30222.7	21703.5	8519.2
Poland	83	112	29549.1	20580.7	8968.4
United States	1714	1751	35165.0	11357.2	23807.8

The original `anchoring.RData` file is kept for optional replication. The required questions can be answered from the tables above.

Answer the following:

1. **Treatment vs. Outcome.** What is the *treatment* in this experiment, and what is the *outcome* variable of interest?

2. **Threats to Internal Validity.** Identify two factors that could undermine internal validity. Think about sample size or potential interference among participants (e.g., if they share information).

3. **Checking randomization.** What would you expect in the data under randomization? Using Table 8, compare the high- and low-anchor groups separately for online and lab participants.

4. **Causal inference under randomization.** Suppose that randomization holds for the lab sample? How might one estimate the *causal* effect of the anchor?

5. **Data Exploration.** Using Table 9, compute the mean difference in `everest_feet` between anchor groups.

6. **Interpretation.** Is there a noticeable difference between the "low" and "high" anchor groups? Compare it to the standard deviation of `everest_feet`.

7. **Causation?** Based on this design, can you say that the anchor *caused* people to give higher or lower estimates? Why or why not?

8. **External Validity.** Comment on threats to the external validity of this study. Use Table 10 to compare the estimated effect across countries.

9. **Optional R Extension:** If you want extra coding practice, reproduce one of the anchoring tables from `anchoring.RData`. This is optional enrichment and is not required for the problem set.

Solution:

1. **Treatment vs. Outcome.**
   The *treatment* is the randomly assigned anchor (`low` or `high`). The *outcome* variable is the participant's numerical estimate of Everest's height in feet (`everest_feet`).

2. **Threats to Internal Validity.**
   Two potential threats include:

   - *Small or unbalanced samples* in the anchor groups, which can increase variance or bias the estimated effect.
   - *Interference among participants,* such as discussing the anchors or looking up Everest's height online, violating the assumption that each individual's treatment does not affect other individuals' outcomes.

3. **Checking Randomization.**
   Under proper randomization, participant characteristics should be similar across the `low` and `high` anchor groups. Table 8 looks reasonably balanced within each sample: mean age is almost identical within the lab sample and very similar within the online sample. Female shares differ only slightly. This does not prove perfect randomization, but it is the kind of pattern we would hope to see.

4. **Causal Inference under Randomization.**
If randomization truly holds for the lab sample, we can estimate the causal effect of the high vs. low anchor by calculating the difference in average `everest_feet` between the high-anchor and low-anchor groups.

   This difference is interpretable as the *causal* effect of being assigned a high anchor (as opposed to a low anchor), provided no major violations of randomization assumptions exist.

5. **Data Exploration.**
In the full sample, the high-anchor group has a mean estimate of 34607.88 feet and the low-anchor group has a mean estimate of 11510.55 feet. The difference is

$$34607.88 - 11510.55 = 23097.33 \text{ feet.}$$

6. **Interpretation.**
The difference is very large. It is more than twice the standard deviation in either anchor group, which suggests that the arbitrary anchor strongly shifted average estimates.

7. **Causation?**
Under *ideal* randomization, the difference in average estimates can be interpreted as *caused* by the anchor. However, if randomization is compromised (especially online), or if participants share information, the difference might also reflect these confounding factors rather than a genuine anchoring effect.

8. **External Validity.**
The estimated effect is positive in every listed country, but its size varies substantially. For example, the difference is about 28619.9 feet in Canada, 23807.8 feet in the United States, 18959.1 feet in India, and only 8519.2 feet in Italy. This variation suggests that the causal effect found in the experiment may not have exactly the same size across populations.

9. **Optional R Extension.**
Optional only: the original `anchoring.RData` file can be used to reproduce the printed tables.

3. Worker Injury Benefits and Time Out of Work

How worker injury benefits affect time out of work is a central question in labor policy. Some argue that generous benefits reduce incentives to return to work, while others see them as promoting fairness and better job matches.

In July 1980, the state of Kentucky raised the maximum weekly worker injury benefit from \$131 to \$217 (**66%** increase). Only high-earning workers (i.e., above a certain income threshold) qualified for the higher benefits, creating a natural experiment for researchers.[2] The file `injury.csv` contains the variables described in Table 11.

[2] Meyer, B.D., W.K. Viscusi, and D.L. Durbin. 1995. Workers' Compensation and Injury Duration: Evidence from a Natural Experiment. *The American Economic Review* 85 (3): 322–340.

Table 11: Description of Variables in the Worker Injury Data

Variable	**Description**
`afchnge`	Equals 1 if the claim occurs after the policy change, 0 otherwise
`ldurat`	Log of the duration (in weeks) out of work
`highearn`	Binary indicator (1 if high wage earner, 0 if low wage earner)

Table 12: Mean log duration out of work by group and period

Group	**Period**	**n**	**Mean `ldurat`**	**SD**
Low earners	Before policy	2294	1.199	1.263
Low earners	After policy	2004	1.223	1.302
High earners	Before policy	1472	1.415	1.320
High earners	After policy	1380	1.627	1.328

The original `injury.RData` file is kept for optional replication. The required questions can be answered from Table 12.

Answer the following questions:

1. Identify the treatment and the outcome variables.
2. Would a Difference-in-Mean estimator between the treated and not-treated groups *after* the policy fairly represent the causal effect in this study? Explain.
3. Suppose you decide to stick to a Before-and-After (BA) design. What would be the treated and control groups in such a case?
4. Using Table 12, describe the four group-period means.
5. Compute the BA estimator and comment on your result.
6. Discuss potential limitations of the BA study, and how the Difference-in-Differences (DiD) design could mitigate them.
7. Why do we need the assumption that, in the absence of the policy, time trends for both groups (high earners and low earners) would have moved in parallel?
8. Recall the DiD estimator:

$$\widehat{\text{SATE}}_{\text{DiD}} = (\overline{Y}_T^{\text{after}} - \overline{Y}_T^{\text{before}}) \; - \; (\overline{Y}_C^{\text{after}} - \overline{Y}_C^{\text{before}}).$$

   Explain, in plain words, the two terms in parentheses.
9. Compute the DiD estimator and comment on the results.
10. List potential factors that might invalidate the parallel trends assumption (e.g., macro shocks hitting the groups differently).

11. **Optional R Extension:** If you want extra coding practice, reproduce Table 12. This is optional enrichment and is not required for the problem set.

Solution:

- **Treatment/Outcome.** Treatment is receiving higher benefits (i.e., being a high earner *after* the policy change). The outcome is `ldurat`, the log duration out of work.
- **Difference-in-Mean After the Policy.** Simply comparing high earners to low earners after the policy overlooks confounders (e.g., work ethics and satisfaction with the workplace) and therefore fails to isolate the policy's causal effect.
- **Before-and-After (BA) Groups.** "Treated" = high earners after policy; "Control" = high earners before policy, if one tries a BA approach.
- **Group-Period Means.** Low earners have mean `ldurat` 1.199 before the policy and 1.223 after the policy. High earners have mean `ldurat` 1.415 before the policy and 1.627 after the policy.
- **BA Estimator.** The BA estimate for high earners is

  $$1.627 - 1.415 = 0.212.$$

  The estimate suggests that log duration out of work rose for high earners after the policy. BA can be biased if other time-varying factors affected high earners at the same time.
- **Limitations and DiD.** BA fails to account for overall time trends or group differences. DiD subtracts out the trend using an untreated "control" group that presumably captures the baseline shift over time.
- **Parallel Trends.** DiD requires that, absent the policy, the outcome trend for high earners would have been parallel to that of low earners. This lets us state that any difference observed after the policy is due to the policy itself.
- **DiD Terms.** $(\overline{Y}_T^{\text{after}} - \overline{Y}_T^{\text{before}})$ is the post-minus-pre difference for the treated group; $(\overline{Y}_C^{\text{after}} - \overline{Y}_C^{\text{before}})$ is the analogous difference for the control group. Their difference isolates the policy's impact, under the parallel-trend assumption.
- **Result of DiD.** The high-earner change is $1.627 - 1.415 = 0.212$. The low-earner change is $1.223 - 1.199 = 0.024$. Therefore,

  $$\widehat{\text{SATE}}_{\text{DiD}} = 0.212 - 0.024 = 0.188.$$

  The DiD estimate suggests that the benefit increase raised log duration out of work by about 0.188 for high earners relative to low earners, assuming parallel trends.
- **Potential Violations.** Any event around July 1980 that unequally affects high- and low-wage workers (e.g., sector-specific shocks) can invalidate parallel trends and bias the DiD estimate.
- **Optional R Extension.** Optional only: the original `injury.RData` file can be used to reproduce Table 12.